Linear Algebra and Its Applications

Section 5.8: Iterative Estimates for Eigenvalues

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.8 Iterative estimates for eigenvalues

Follow the book's indexing carefully. Starting with x_0, compute $y_k = Ax_k$, choose an entry μ_k of y_k with largest absolute value (retain its sign), and set $x_{k+1} = y_k/\mu_k$. Thus μ_1 comes from Ax_1, not Ax_0. A negative dominant eigenvalue need not cause the normalized vectors to alternate, because the signed normalization removes the reversal.

For inverse iteration with a shift α, solve $(A-\alpha I)y_k = x_k$ instead. Normalize by its largest-magnitude entry μ_k, and estimate the original eigenvalue by $\theta_k = \alpha + 1/\mu_k$. Reuse a factorization of $A - \alpha I$; do not explicitly form its inverse at every step. A small residual $\|Av - \theta v\|$ checks a proposed eigenpair. Agreement of two consecutive estimates alone can be misleading.

Practice problem 5.8.P1: Test proportionality and inspect the residual

The given vector $x = (1, -4.3, 8.1)^T$ gives $Ax = (3, -13, 24.5)^T$. The entrywise ratios are 3, $130/43 \approx 3.023256$, and $245/81 \approx 3.024691$. Their close agreement suggests an eigenvalue near 3.024. The Rayleigh estimate is

$$R(x) = \frac{x^T Ax}{x^T x} = \frac{25735}{8510} \approx 3.024089.$$

For $\theta = R(x)$, inspect $Ax - \theta x$ relative to the size of Ax or x. A residual near zero, rather than the raw size of x, is the meaningful approximation test. Exact proportionality would give zero residual and an exact eigenvector.

Exercise 5.8.1: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} 1 \\ \frac{1663}{5000} \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} \frac{24989}{5000} \\ \frac{4163}{2500} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx 4.997800$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} 1 \\ \frac{1}{3} \end{bmatrix}$, and direct multiplication checks $Av = 5v$ exactly.

Conclusion. The dominant eigenvalue is 5, with eigenvector $\begin{bmatrix} 1 \\ \frac{1}{3} \end{bmatrix}$.

Exercise 5.8.2: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} -\frac{63}{250} \\ 1 \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} -\frac{1567}{1250} \\ \frac{3129}{625} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx 5.006400$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} -\frac{1}{4} \\ 1 \end{bmatrix}$, and direct multiplication checks $Av = 5v$ exactly.

Conclusion. The dominant eigenvalue is 5, with eigenvector $\begin{bmatrix} -\frac{1}{4} \\ 1 \end{bmatrix}$.

Exercise 5.8.3: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} \frac{1297}{2500} \\ 1 \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} \frac{2297}{5000} \\ \frac{2836}{3125} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx 0.907520$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$, and direct multiplication checks $Av = \frac{9}{10}v$ exactly.

Conclusion. The dominant eigenvalue is $\frac{9}{10}$, with eigenvector $\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$.

Exercise 5.8.4: Read the limiting direction and retain the scaling sign

The final supplied normalized vector is approximately $\begin{bmatrix} 1 \\ \frac{3751}{5000} \end{bmatrix}$. Multiplication by the stated matrix gives

$$Ax \approx \begin{bmatrix} -\frac{1003}{2500} \\ -\frac{3761}{12500} \end{bmatrix}.$$

The largest-magnitude component is $\mu \approx -0.401200$. The trend in the vectors suggests the sharper limiting direction $v = \begin{bmatrix} 1 \\ \frac{3}{4} \end{bmatrix}$, and direct multiplication checks $Av = -\frac{2}{5}v$ exactly.

Conclusion. The dominant eigenvalue is $-\frac{2}{5}$, with eigenvector $\begin{bmatrix} 1 \\ \frac{3}{4} \end{bmatrix}$.

Here "largest" in the exercise means largest in *absolute value*: the other eigenvalue is .1, which is algebraically larger but not dominant.

Exercise 5.8.5: Normalize by the second component of the supplied iterate

Dividing the last vector by its second entry gives

$$x = \begin{bmatrix} -\frac{24991}{31241} \\ 1 \end{bmatrix} \approx \begin{bmatrix} -\frac{7999}{10000} \\ 1 \end{bmatrix}.$$

Use the requested four-decimal vector for a check:

$$A \begin{bmatrix} -\frac{7999}{10000} \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{8003}{2000} \\ -\frac{2501}{500} \end{bmatrix}.$$

The second entry estimates the eigenvalue by $-\frac{2501}{500}$, since the corresponding input entry is 1. The nearby exact eigenpair is $\lambda = -5$, $v = \begin{bmatrix} -\frac{4}{5} \\ 1 \end{bmatrix}$, verified by direct multiplication. The rounded estimate is approximate, so exact equality $Ax = \lambda x$ is not expected until using the exact vector.

Exercise 5.8.6: Normalize by the second component of the supplied iterate

Dividing the last vector by its second entry gives

$$x = \begin{bmatrix} -\frac{2045}{4093} \\ 1 \end{bmatrix} \approx \begin{bmatrix} -\frac{1249}{2500} \\ 1 \end{bmatrix}.$$

Use the requested four-decimal vector for a check:

$$A \begin{bmatrix} -\frac{1249}{2500} \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{2501}{1250} \\ \frac{5003}{1250} \end{bmatrix}.$$

The second entry estimates the eigenvalue by $\frac{5003}{1250}$, since the corresponding input entry is 1. The nearby exact eigenpair is $\lambda = 4$, $v = \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix}$, verified by direct multiplication. The rounded estimate is approximate, so exact equality $Ax = \lambda x$ is not expected until using the exact vector.

Exercise 5.8.7: Compute the requested five indexed power-method rows

Use the signed largest-magnitude entry for normalization and do not round intermediate vectors. For $A = \begin{bmatrix} 6 & 7 \\ 8 & 5 \end{bmatrix}$ and $x_0 = (1, 0)^T$, the results are

k	$(x_k)_1$	$(x_k)_2$	μ_k
1	0.750000000	1.000000000	11.500000000
2	1.000000000	0.956521739	12.782608696
3	0.993197279	1.000000000	12.959183673
4	1.000000000	0.998950131	12.994750656
5	0.999838416	1.000000000	12.999030499

.

For an indexing check, the first multiplication gives $Ax_0 = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$ and $\mu_0 = 8$. Dividing by this scalar produces x_1, after which μ_1 is computed from Ax_1. The values converge toward the dominant eigenvalue 13, with eigenvector direction $(1, 1)^T$.

Exercise 5.8.8: Compute the requested five indexed power-method rows

Use the signed largest-magnitude entry for normalization and do not round intermediate vectors. For $A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$ and $x_0 = (1, 0)^T$, the results are

k	$(x_k)_1$	$(x_k)_2$	μ_k
1	0.500000000	1.000000000	7.000000000
2	0.285714286	1.000000000	6.142857143
3	0.255813953	1.000000000	6.023255814
4	0.250965251	1.000000000	6.003861004
5	0.250160772	1.000000000	6.000643087

.

For an indexing check, the first multiplication gives $Ax_0 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ and $\mu_0 = 4$. Dividing by this scalar produces x_1, after which μ_1 is computed from Ax_1. The values converge toward the dominant eigenvalue 6, with eigenvector direction $(1/4, 1)^T$.

Exercise 5.8.9: Continue the same indexed algorithm to the sixth estimate

For the given matrix

$$A = \begin{bmatrix} 8 & 0 & 12 \\ 1 & -2 & 1 \\ 0 & 3 & 0 \end{bmatrix}, \qquad x_0 = (1, 0, 0)^T,$$

repeatedly compute Ax_k, retain its signed largest-magnitude entry, and divide to form x_{k+1}. This gives

$$\mu_5 \approx 8.4233109850, \qquad \mu_6 \approx 8.4246301920.$$

The accompanying current vectors, which make these estimates reproducible, are

$$x_5 \approx \begin{bmatrix} 1.0 \\ 0.0993553378 \\ 0.0352759154 \end{bmatrix}, \qquad x_6 \approx \begin{bmatrix} 1.0 \\ 0.0993154879 \\ 0.0353858493 \end{bmatrix}.$$

Multiplying each vector by A and selecting its largest-magnitude entry checks the listed scalars. The vectors have been rounded only for display, so the check agrees to the displayed precision.

Exercise 5.8.10: Continue the same indexed algorithm to the sixth estimate

For the given matrix

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 9 \\ 0 & 1 & 9 \end{bmatrix}, \qquad x_0 = (1, 0, 0)^T,$$

repeatedly compute Ax_k, retain its signed largest-magnitude entry, and divide to form x_{k+1}. This gives

$$\mu_5 \approx 9.9319371728, \qquad \mu_6 \approx 9.9871727289.$$

The accompanying current vectors, which make these estimates reproducible, are

$$x_5 \approx \begin{bmatrix} 0.0183246073 \\ 1.0 \\ 0.9904013962 \end{bmatrix}, \qquad x_6 \approx \begin{bmatrix} 0.0037778949 \\ 1.0 \\ 0.9981549815 \end{bmatrix}.$$

Multiplying each vector by A and selecting its largest-magnitude entry checks the listed scalars. The vectors have been rounded only for display, so the check agrees to the displayed precision.

Exercise 5.8.11: Compare the scaling estimate with the Rayleigh quotient

The exact dominant eigenvalue, checked from the characteristic polynomial, is 6. Compute each Rayleigh quotient from the current x_k using $R(x_k) = x_k^T A x_k / (x_k^T x_k)$:

k	μ_k	$R(x_k)$	$\lvert\mu_k - \lambda\rvert$	$\lvert R(x_k) - \lambda\rvert$
1	5.800000000000	5.965517241379	$2.000e-01$	$3.448e-02$
2	5.965517241379	5.999035679846	$3.448e-02$	$9.643e-04$
3	5.994219653179	5.999973208305	$5.780e-03$	$2.679e-05$
4	5.999035679846	5.999999255782	$9.643e-04$	$7.442e-07$

.

The quotient's error is substantially smaller at the same iterate. Both matrices are symmetric, so this demonstrates the improved accuracy claimed in the exercise. Signed normalization is especially important for the negative dominant eigenvalue in Exercise 12.

Exercise 5.8.12: Compare the scaling estimate with the Rayleigh quotient

The exact dominant eigenvalue, checked from the characteristic polynomial, is -4. Compute each Rayleigh quotient from the current x_k using $R(x_k) = x_k^T A x_k / (x_k^T x_k)$:

k	μ_k	$R(x_k)$	$\lvert\mu_k - \lambda\rvert$	$\lvert R(x_k) - \lambda\rvert$
1	-4.333333333333	-3.923076923077	$3.333e-01$	$7.692e-02$
2	-3.923076923077	-3.995121951220	$7.692e-02$	$4.878e-03$
3	-4.019607843137	-3.999694842844	$1.961e-02$	$3.052e-04$
4	-3.995121951220	-3.999980926586	$4.878e-03$	$1.907e-05$

.

The quotient's error is substantially smaller at the same iterate. Both matrices are symmetric, so this demonstrates the improved accuracy claimed in the exercise. Signed normalization is especially important for the negative dominant eigenvalue in Exercise 12.

Exercise 5.8.13: A theoretically convergent method can be impractically slow

If the two leading eigenvalues have different moduli, there is a strictly dominant one, so the power method converges for an initial vector with a nonzero component in that eigendirection (under the section's eigenbasis hypotheses). But the ratio of the other large modulus to the dominant modulus is near $4/4 = 1$. The unwanted mode therefore decays slowly relative to the desired one.

Conclusion. It can work, but may require many iterations and be less useful than shifted inverse iteration. An initial vector with no dominant component is an exception to the usual convergence claim.

Exercise 5.8.14: Break the tie by shifting toward the positive eigenvalue

If the leading eigenvalues are exactly λ and $-\lambda$, their powers have equal moduli; ordinary power iteration need not select a single direction. Choose a shift α near the positive eigenvalue (for example 4.1, if suitably close) but not exactly an eigenvalue. Inverse iteration applies to $(A-\alpha I)^{-1}$. Its eigenvalues are $1/(\lambda_j-\alpha)$, so the desired positive eigenvalue is dominant when its distance from the shift is uniquely smallest. Solving shifted systems and normalizing then gives a sequence for that positive eigenpair. Alternatively, adding a suitable positive multiple of I can create a unique positive dominant modulus; subtract that shift from the resulting eigenvalue estimate. The inverse method is usually faster when an accurate initial estimate is available.

Exercise 5.8.15: Transform an eigenvalue under a shifted inverse

From $Ax=\lambda x$ subtract αx to get $(A-\alpha I)x=(\lambda-\alpha)x$. Since α is not an eigenvalue, the matrix is invertible and $\lambda-\alpha\neq 0$. Multiply by $(A-\alpha I)^{-1}$:

$$x=(\lambda-\alpha)(A-\alpha I)^{-1}x, \qquad (A-\alpha I)^{-1}x=\frac{1}{\lambda-\alpha}x.$$

Because $x\neq 0$, this is exactly the claimed eigenpair.

Exercise 5.8.16: Recover the original eigenvalue from the inverse one

Multiply $(A-\alpha I)^{-1}x=\mu x$ by $A-\alpha I$: $x=\mu(A-\alpha I)x$. The inverse is invertible, so its eigenvalue μ is nonzero. Dividing by μ and rearranging gives

$$Ax=\left(\alpha+\frac{1}{\mu}\right)x.$$

Thus $\alpha+1/\mu$ is an eigenvalue of A, with the same nonzero eigenvector. This proves the scalar update in inverse iteration.

Exercise 5.8.17: Run shifted inverse iteration and verify its residual

Use

$$A=\begin{bmatrix}10&-8&-4\\-8&13&4\\-4&5&4\end{bmatrix}, \quad \alpha=3.3, \quad x_0=(1,0,0)^T.$$

At each step solve $(A-\alpha I)y_k=x_k$, choose a signed largest-magnitude component μ_k, and set $x_{k+1}=y_k/\mu_k$, $\theta_k=\alpha+1/\mu_k$. Using high-precision arithmetic, iteration index $k=7$ gives

$$\theta_k\approx 3.3212201246571, \qquad x_{k+1}\approx\begin{bmatrix}1.0\\0.787012129501\\0.0956707098337\end{bmatrix}.$$

Before rounding, $\|Ax_{k+1}-\theta_k x_{k+1}\|_\infty<10^{-10}$ and the last two scalar estimates differ by less than 10^{-12}. The eigenvalue to four decimal places is $\boxed{3.3212}$. Independent roots of the exact characteristic polynomial confirm this choice. The shift makes this eigenvalue uniquely closest, so inverse iteration selects it rather than the dominant eigenvalue of A.

Exercise 5.8.18: Run shifted inverse iteration and verify its residual

Use

$$A = \begin{bmatrix} 8 & 0 & 12 \\ 1 & -2 & 1 \\ 0 & 3 & 0 \end{bmatrix}, \quad \alpha = -1.4, \quad x_0 = (1, 0, 0)^T.$$

At each step solve $(A - \alpha I)y_k = x_k$, choose a signed largest-magnitude component μ_k, and set $x_{k+1} = y_k/\mu_k$, $\theta_k = \alpha + 1/\mu_k$. Using high-precision arithmetic, iteration index $k = 10$ gives

$$\theta_k \approx -1.424428900898, \qquad x_{k+1} \approx \begin{bmatrix} 1.0 \\ 0.372900802803 \\ -0.785369075075 \end{bmatrix}.$$

Before rounding, $\|Ax_{k+1} - \theta_k x_{k+1}\|_\infty < 10^{-10}$ and the last two scalar estimates differ by less than 10^{-12}. The eigenvalue to four decimal places is $\boxed{-1.4244}$. Independent roots of the exact characteristic polynomial confirm this choice. The shift makes this eigenvalue uniquely closest, so inverse iteration selects it rather than the dominant eigenvalue of A.

Exercise 5.8.19: Compute both extreme-modulus eigenpairs from the required start

Use $x_0 = (1, 0, 0, 0)^T$ for both computations.

(a) Ordinary power iteration.

With signed largest-component normalization, the scalar estimate converges to

$$\lambda_{\max} \approx 30.288685345802, \quad v_{\max} \approx \begin{bmatrix} 0.95762878047 \\ 0.68893692056 \\ 1.0 \\ 0.94378150874 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{30.2887}$.

(b) Inverse iteration with shift zero.

Repeatedly solve $Ay_k = x_k$ instead of explicitly multiplying by A^{-1}. The smallest-modulus eigenpair is

$$\lambda_{\text{near }0} \approx 0.010150048397894, \quad v_{\text{near }0} \approx \begin{bmatrix} -0.60397234234 \\ 1.0 \\ -0.25113513051 \\ 0.14895344556 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{0.0102}$. The stopping tests were both a scalar-estimate change below 10^{-12} and an unrounded residual below 10^{-10}. They hold at indices $k = 16$ and $k = 6$, respectively, with $k = 0$ representing the first multiplication or solve. Independent characteristic-polynomial roots verify which eigenvalues are largest and nearest zero. The displayed eigenvectors are scaled to make their largest-magnitude component equal to 1.

Exercise 5.8.20: Compute both extreme-modulus eigenpairs from the required start

Use $x_0 = (1, 0, 0, 0)^T$ for both computations.

(a) Ordinary power iteration.

With signed largest-component normalization, the scalar estimate converges to

$$\lambda_{\max} \approx 19.182036763332, \quad v_{\max} \approx \begin{bmatrix} 0.1844163575 \\ 1.0 \\ 0.17961525893 \\ 0.40710960748 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{19.1820}$.

(b) Inverse iteration with shift zero.

Repeatedly solve $Ay_k = x_k$ instead of explicitly multiplying by A^{-1}. The smallest-modulus eigenpair is

$$\lambda_{\text{near } 0} \approx 0.012205562828844, \quad v_{\text{near } 0} \approx \begin{bmatrix} 1.0 \\ 0.22260978915 \\ -0.91799334087 \\ 0.66048300357 \end{bmatrix}.$$

The four-decimal eigenvalue is $\boxed{0.0122}$. The stopping tests were both a scalar-estimate change below 10^{-12} and an unrounded residual below 10^{-10}. They hold at indices $k = 18$ and $k = 6$, respectively, with $k = 0$ representing the first multiplication or solve. Independent characteristic-polynomial roots verify which eigenvalues are largest and nearest zero. The displayed eigenvectors are scaled to make their largest-magnitude component equal to 1.

Exercise 5.8.21: Separate vector size from convergence of direction

For $x = (.5, .5)^T$, diagonal multiplication is explicit:

(a) $A^k x = (.5(.8)^k, .5(.2)^k)^T \to 0$. The zero vector is not an eigenvector. Yet the second coordinate relative to the first is $(1/4)^k \to 0$, so the direction tends to the first coordinate axis.

(b) $A^k x = (.5, .5(.8)^k)^T \to (.5, 0)^T$, a nonzero eigenvector. Here the dominant eigenvalue is exactly 1, so its scale is preserved.

(c) $A^k x = (.5(8)^k, .5(2)^k)^T$ grows without bound. Its direction again tends to the first coordinate axis, but it does not converge to a finite vector.

The common conclusion is about normalized direction, not convergence of $A^k x$ itself. Moreover, a vector exactly missing the dominant mode cannot reveal that eigendirection. Appropriate normalization is therefore an essential part of the power method, not an optional cosmetic step.

Reproducing the iterative computations. The following complete functions implement the signed indexing used in the tables. Enter the matrix printed in the chosen exercise and its stated initial vector as NumPy arrays. For 19–20 and inverse iteration, continue until both the estimate change and eigenpair residual are small.

```
import numpy as np
def power_rows(A, x, count):
    for k in range(count+1):
        y = A @ x
        mu = y[np.argmax(np.abs(y))]
        rayleigh = (x @ A @ x)/(x @ x)
        yield k, x.copy(), mu, rayleigh
        x = y/mu

def inverse_rows(A, alpha, x, count):
    H = A-alpha*np.eye(len(x))
    for k in range(count+1):
        y = np.linalg.solve(H, x)
        mu = y[np.argmax(np.abs(y))]
        xnew = y/mu
        theta = alpha+1/mu
        residual = np.max(np.abs(A@xnew-theta*xnew))
        yield k, xnew.copy(), theta, residual
        x = xnew
```

All numerical results above were actually computed from the supplied matrices; the code is provided to reproduce them. Extra working precision was used for the final residual checks in Exercises 17–20.